

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology and Geophysics

2. Subject Name: Igneous Petrology Lab L-T-P: 0-0-3 Credits: 2

3. Pre-requisites: None

4. Syllabus and reference books:

Petrographic study of a wide spectrum of igneous rocks in thin sections; Calculation of normative mineralogy (NORM); Use of ternary phase diagrams to trace/track crystallization behavior of magmas; Modelling partial melting and fractional crystallization processes using mass balance models and mineral-melt partition coefficients

Reference Books:

- 1) Atlas of igneous rocks and their textures by W.S. MacKenzie, C.H. Donaldson and C. Guilford
- 2) Principles of Igneous and Metamorphic Petrology by John D. Winter
- 3) Using Geochemical Data: evaluation, presentation, interpretation by H. R. Rollinson

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Petrographic study of a wide spectrum of igneous rocks in thin sections	6
2.	Calculation of normative mineralogy (NORM)	2
3.	Use of ternary phase diagrams to trace/track crystallization behavior of magmas	2
4.	Modelling partial melting and fractional crystallization processes using mass balance models and mineral-melt partition coefficients	2
Total number of hours		12*3 = 36